BEGINNER-FRIENDLY TECHNICAL REVIEW

Could this 337 nm pulse be detected at 20 km?

FEGS on NASA's ER-2 and an ASIM-like photometer

Rechecked against primary instrument papers and the supplied figure.

Main conclusion

Taking the plotted mW/m^2 values literally, the pulse contains ample light for an appropriately configured ASIM-class detector. An unmodified ASIM photon-counting channel would be overloaded if it accepted most of the plotted peak. FEGS blue-channel detection is **plausible and likely on brightness grounds**, but is not established by a verified absolute blue-channel threshold.
Brightness alone does not establish detection at a particular offset. The light must also arrive from directions the instrument can see, and the cloud-transport calculation must have the correct absolute normalization.

Reference case: aircraft at 20 km above mean sea level; compact light-emitting cloud-exit patch at 15 km; level, downward-pointing instruments. The nominal geometry gives the following *horizontal offsets from the point directly below the aircraft*, not slant distances:

	Toward a side	Toward a corner
FEGS 337 nm channel	0.83 km	1.18 km
ASIM-like 337 nm channel	2.97 km	4.20 km

These are calculated compact-patch viewing limits, not measured detection ranges. Instrument geometry comes from Refs. [1, 2, 3, 4].

A compact cloud-exit patch **5 km ahead is outside both footprints in this 15 km case**. A buried discharge at that offset could nevertheless illuminate a scattered cloud halo inside the footprint. The time trace alone does not tell us whether that halo is bright enough.

What the double-check changed or clarified

A new normalization flag: rough integration of the image gives about three to four times more fluence in the dark scattered curve than in the light unscattered curve. Temporal broadening alone cannot explain that increase; spatial/angular redistribution or another explicit model factor would be needed.

A stronger hardware warning: the ASIM instrument paper identifies a daylight damage risk, not merely a noisy daytime background. Its original flight hardware should not be treated as a daytime-ready airborne sensor.[1]

A narrower FEGS claim: the earlier red-channel sensitivity comparison is not a calibration of the 337 nm channel. It is retained only as a labelled comparison in Appendix A.

Scope. This is a conditional detectability assessment, not a validation of the discharge model, a certified instrument operating prescription, or a new three-dimensional cloud simulation. Pages 2–8 explain the result; page 9 gives the calculation needed for a true detection-versus-offset map.

1 Read the supplied plot before comparing detectors

The relevant curve is the **dark-blue scattered 337.1 nm trace in the bottom panel**. The much taller light-blue trace is the unscattered comparison. They must not be interchanged.

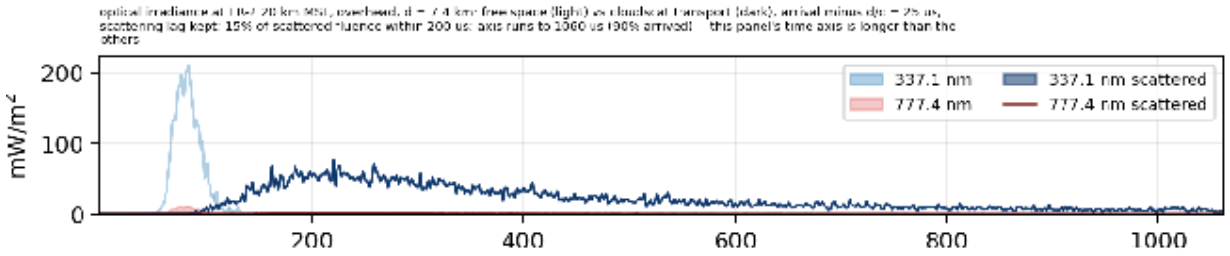


Figure 1: The supplied optical panel, cropped without changing its values. Small labels retain the resolution of the original image. The caption indicates a time axis in microseconds extending to about 1060 μs.

Two quantities describe the strength of a pulse

Irradiance, $E(t)$, is optical power arriving per unit collecting area. Its unit is W/m^2. The plotted unit is mW/m^2, so a value of 50 means

$$50\,\mathrm{mW\,m^{-2}} = 0.050\,\mathrm{W\,m^{-2}}.$$

This describes how strong the pulse is at an instant. It is not the total power emitted by the source.

Fluence, H, is the energy arriving per unit area over the pulse. It is the area under the irradiance–time curve:

$$H = \int E(t)\,dt \qquad [\mathrm{J\,m^{-2}}]. \tag{1}$$

A narrow tall pulse and a broad low pulse can have the same fluence. Fluence is especially useful when the detector combines several time samples.

Image-based quantity	Value used in this review
Scattered peak envelope	Approximately 50–70 mW/m^2
Representative peak for calculations	$E_{\mathrm{ref,pk}} = 50$ mW/m^2
Scattered fluence in the displayed interval	$H_{\mathrm{ref}} \simeq 2 \times 10^{-5}$ J/m^2
Unscattered fluence, rough comparison	About 6×10^{-6} J/m^2
Displayed scattered duration	Of order 1 ms; 1 ms = 1000 μs

These are rounded estimates from a raster image, not the original simulated samples. The image is not precise enough to justify small percentage differences. Its caption appears to place roughly 90% of the scattered arrival distribution inside the displayed window; the full tail has not been recovered here.

Do not apply the distance factor twice

The bottom panel is labelled as irradiance at the ER-2. If that is its true meaning, geometrical dilution and any cloud loss already included in the model must *not* be applied again. A new aircraft offset, however, generally requires a new transport/geometry calculation.

2 An important check on the scattering normalization

Scattering redirects light and changes arrival times. A longer pulse is therefore reasonable. But a delay model is not automatically a complete model of how much light a detector receives.

What a time-broadening model can and cannot do

Suppose the unscattered reference is $E_0(t)$ and a delay distribution is $K(t)$. A model that *only* spreads arrivals in time can be written

$$E_{\text{broad}}(t) = \int E_0(t - t')K(t')\,dt', \qquad \int K(t')\,dt' = 1. \tag{2}$$

The second condition means that the delay probabilities add to one. Integrating the first expression over all time gives

$$\int E_{\text{broad}}(t)\,dt = \int E_0(t)\,dt. \tag{3}$$

In ordinary language, **rescheduling the same arriving photons does not create extra arriving energy**.

The rough image integrals instead give

$$\frac{H_{\text{scattered,shown}}}{H_{\text{unscattered}}} \simeq \frac{2 \times 10^{-5}}{6 \times 10^{-6}} \simeq 3.3.$$

The exact ratio needs the numerical arrays, but the apparent difference is large enough to check in the model code.

This is a flag, not proof that the transport is unphysical

Real scattering can redirect photons *toward a particular observer*, so the received fluence at one location can exceed its free-space comparison without violating total energy conservation. The issue is whether the model explicitly and correctly includes that spatial/angular redistribution. A unit-area temporal convolution by itself would not explain the plotted increase.

What must be verified in the simulation

Confirm that both curves use the same source normalization, spectral band, units, collecting plane, and meaning of "irradiance." Then distinguish a prescribed delay kernel from a physically normalized cloud calculation. Three-dimensional photon-transport models can account for cloud geometry, scattering and the observer's response.[6]

The title refers to an EFCM source 5 km ahead, while the optical caption says "overhead" and gives a distance near 7.4 km. These may intentionally describe different instrument geometries. **The optical curve should not be assumed to already represent the 5 km-offset case.**

Assumptions for the following photon calculation

The plotted signal is treated as band-integrated 337 nm irradiance at the aircraft, rather than power per nanometre or radiance per steradian. Initially, all of that reference light is assumed to be accepted by the photometer. Angular rejection, extra window losses and additional attenuation are introduced separately, only when not already included.

Even reducing the plotted amplitude by a factor of three or four would leave a very large ideal ASIM photon input under full acceptance. That robustness does not validate the model's absolute normalization; it only shows that this particular factor would not erase the sensitivity margin.

3 ASIM-like sensor: count the available photons

A photometer measures light versus time rather than forming a detailed image. The ASIM/MMIA 337 nm sensor converts some incident photons into electrical pulses. A **photoelectron** is an electron released when the detector responds to a photon.

The following nominal inputs are from the instrument description, Tables 2 and 5.[1]

Parameter	337 nm photometer input
Aperture diameter	1.0 mm
In-band efficiency, including quantum efficiency	$\eta = 0.136$
Spectral band / width	337 nm / 4 nm
Sample duration	$\Delta t = 10\ \mu$s
Field of view	Square, 80° full diagonal
Maximum design count-rate scale	100 MHz = 10^8 s^{-1}

Step 1: energy carried by one photon

For wavelength $\lambda = 337.1$ nm,

$$\epsilon_\gamma = \frac{h_P c}{\lambda} = 5.89 \times 10^{-19}\ \text{J}. \tag{4}$$

Here h_P is Planck's constant and c is the speed of light. The subscript prevents confusion with the aircraft–cloud clearance h used later.

Step 2: effective collecting area

The physical opening has area $A = \pi(0.001/2)^2 = 7.85 \times 10^{-7}$ m^2. Multiplying by the detection efficiency gives

$$A_{\text{eff}} = A\eta = 1.07 \times 10^{-7}\ \text{m}^2. \tag{5}$$

This is a convenient way to account for photons lost in the instrument. **Do not multiply by quantum efficiency a second time**; it is already included in η.

Step 3: predicted input during one sample

Using the representative accepted peak of 0.050 W/m^2,

$$N_{\text{pe,bin}} = \frac{E_{\text{ref,pk}} A_{\text{eff}} \Delta t}{\epsilon_\gamma} \simeq 9.1 \times 10^4 \text{ photoelectrons per } 10\ \mu\text{s}. \tag{6}$$

The displayed pulse fluence similarly gives approximately 3.6×10^6 photoelectrons over the event, *before* saturation or added losses.

How to interpret these large numbers

They are ideal photon-input estimates, **not recorded counts from overloaded hardware**. They show that a lack of photons is not the main problem when the plotted light is accepted. The next questions are operating range, background and field of view.

4 ASIM-like sensor: sensitivity is not operating range

Too much light can spoil a measurement

The ideal peak rate is $R_{\text{ideal}} \simeq 9.1 \times 10^9\ \text{s}^{-1}$. That is about 91 times the design count-rate scale in the preceding table. At high rates, electrical pulses overlap; the paper describes pile-up and saturation.[1]

The irradiance corresponding to 10^8 ideal photoelectrons per second is

$$E_{100\,\text{MHz}} = \frac{10^8 \epsilon_\gamma}{A_{\text{eff}}} \simeq 0.55\,\text{mW}\,\text{m}^{-2}. \tag{7}$$

Thus, even about 1.1% of the reference peak reaches this scale. Being just below it does not guarantee linear, undistorted measurements.

Illustrative attenuation only. Dividing the incoming signal by a factor D changes the ideal input as follows. These calculations do not specify a daylight-safe hardware design.

D	Peak per 10 μs	Pulse total	Interpretation
1	91,000	3.6×10^6	Severe overload expected
100	910	36,000	Near the maximum-rate scale
1,000	91	3,600	More headroom; pile-up still requires checking
10,000	9.1	360	Lower input rate; noisier individual samples

A separate sensitivity comparison

Table 6 of the MMIA paper gives a lower sprite/lightning sensitivity scale of 1.3×10^5 photons per cm^2 per event, using a 3 ms photometer event assumption.[1] Converting it to energy per area gives

$$H_{\text{bench}} = (1.3 \times 10^5)(10^4)\epsilon_\gamma = 7.7 \times 10^{-10}\,\text{J}\,\text{m}^{-2}.$$

The reference fluence is about 2.6×10^4 times larger. **This is an event-sensitivity benchmark, not a universal trigger threshold** for every duration, background or airborne implementation.

Daylight requires a different engineering assessment

The flight-instrument paper warns that powered photometers can be permanently damaged after seconds of daylight exposure and describes a daylight interlock.[1] An airborne "ASIM-like" detector therefore needs its own background protection, gain/readout design and calibration. Attenuation factors calculated from the transient alone do not establish safe daytime operation.

An overload may produce a distorted signal, not a usable pulse measurement. A suitably adapted sensor could detect the transient readily under adequate acceptance and background conditions, but normal triggering, amplitude recovery and fine waveform reconstruction must be evaluated separately.

5 FEGS: what is known, and what remains conditional

The ALOFT configuration includes a nadir-directed photometer centred at 340 nm with a 10 nm full-width-at-half-maximum passband, covering the 337.1 nm emission. Its sampling interval is 10 μs.[3] "Nadir" means directly downward. "Full width at half maximum" describes the width of a filter response at half of its peak transmission.

The sampling is fast enough to represent the broad structure of the supplied millisecond-scale pulse. It does not by itself guarantee adequate sensitivity or freedom from saturation.

The blue channel is not the whole FEGS array

The main 25-channel 777 nm array forms a large combined footprint. The additional spectral channels, including the blue/near-ultraviolet channel, look at the central footprint, nominally about 2×2 km for a 15 km cloud top.[4] The detailed angular model gives about 1.7×1.7 km at 5 km clearance; the nominal description is rounded.

A key distinction for the 5 km-ahead question

The full array's approximately 10×10 km coverage must not be assigned to the 337 nm channel. A red-channel detection near an outer array edge does not establish that the blue channel can directly see the same compact patch.

Why the detection conclusion is still conditional

FEGS has demonstrated 337 nm observations over a sunlit cloud. The 2025 campaign paper also describes radiometric calibration work then in progress.[5] This establishes practical capability, but not an absolute threshold for the particular plotted event.

An applicable absolute 337 nm calibration, noise level and saturation limit were not verified in the sources reviewed. Consequently, the supported judgement is **likely on brightness grounds, not quantitatively certified**. Appendix A checks the earlier 777 nm comparison and explains why it cannot be promoted to a blue-channel threshold.

A simple way to test an actual blue-channel dataset

Let C_{337} convert accepted irradiance into the digitizer's signal units, and let σ_{bg} be the root-mean-square background fluctuation in those units. An illustrative six-standard-deviation test is

$$C_{337}\, E_{\mathrm{acc,pk}} \;\geq\; 6\sigma_{\mathrm{bg}}. \tag{8}$$

The factor six is an example detection rule, not a newly verified blue-channel requirement. The pulse must also stay within the calibrated operating range.

For full acceptance of a 50 mW/m^2 peak, that example test requires an equivalent irradiance noise below $50/6 \simeq 8.3$ mW/m^2. This is a *condition to check*, not a measured FEGS noise value. Filtering, the time window and background estimation all affect the comparison.

Finally, a wider filter does not automatically multiply a narrow emission-band signal by its bandwidth ratio. What matters is transmission at the actual emission wavelengths, together with the extra background admitted by that filter.

6 Understand the field of view and the meaning of offset

Imagine looking at a cloud through a square window. A very bright compact patch is still excluded if it is outside that window. A photometer's **field of view (FOV)** is its accepted range of viewing directions; its **footprint** is where those directions intersect the cloud.

Use the height above the visible light-exit layer. Let z_c be the height from which the photons finally emerge toward the observer. For an aircraft at 20 km,

$$h = 20 - z_c \quad \text{(km)}. \tag{9}$$

The buried discharge altitude need not be z_c. All heights here are measured above mean sea level. The geometrical model assumes a flat exit layer and no further angular redistribution on the path above it.

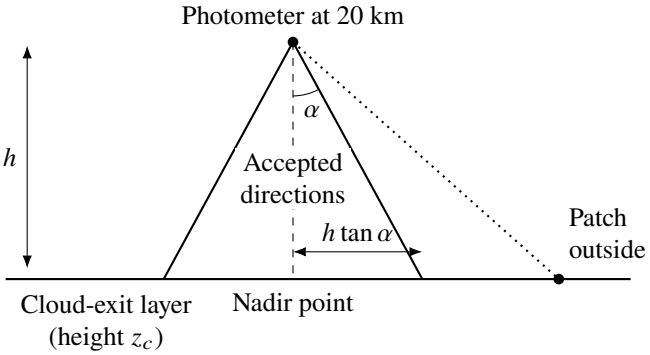


Figure 2: Side view of an idealized field of view. This is an explanatory schematic, not a reconstruction of the cloud or the instrument's measured angular response.

A square has two different limiting radii

If α is the *side* half-angle, the footprint extends to $h \tan \alpha$ on each side of the nadir point. Its corner is farther away by a factor of $\sqrt{2}$:

$$\rho_{\text{side}} = h \tan \alpha, \qquad \rho_{\text{corner}} = \sqrt{2}\, h \tan \alpha. \tag{10}$$

The FEGS single-head side half-angle is 9.46°.[2] ASIM's 80° specification is the *full diagonal*, not the side width.[1] Therefore its corner half-angle is 40° and its side half-angle is

$$\alpha_{\text{ASIM}} = \arctan\left(\frac{\tan 40^\circ}{\sqrt{2}}\right) \simeq 30.68^\circ.$$

A compact patch is inside at every azimuth when $\rho \le \rho_{\text{side}}$, inside only at some azimuths between the two radii, and outside when $\rho > \rho_{\text{corner}}$. Near an edge, real vignetting and aircraft attitude still matter.

7 Calculated offsets, including the 5 km case

For the idealized downward-pointing geometry,

$$\begin{aligned} \text{FEGS}: &\quad \rho_{\text{side}} = 0.1666h, \quad \rho_{\text{corner}} = 0.2356h, \\ \text{ASIM-like}: &\quad \rho_{\text{side}} = 0.5933h, \quad \rho_{\text{corner}} = 0.8391h. \end{aligned}$$

All distances in each equation use the same unit. The cloud-exit heights below are scenarios, *not heights inferred from the supplied plot*.

Exit height	Clearance h	FEGS side–corner	ASIM side–corner
12 km	8 km	1.33–1.89 km	4.75–6.71 km
15 km	5 km	0.83–1.18 km	2.97–4.20 km
17 km	3 km	0.50–0.71 km	1.78–2.52 km
18 km	2 km	0.33–0.47 km	1.19–1.68 km

A patch 5 km from nadir

Exit at 15 km: both blue-channel footprints exclude a compact patch at 5 km, even toward a corner. The viewing angle is $\arctan(5/5) = 45°$, outside even ASIM's 40° diagonal half-angle.

Exit at 12 km: FEGS still excludes it. ASIM-like optics can accept it near a diagonal, but not along a side axis. The actual aircraft heading and instrument orientation determine which applies.

Exit at 17–18 km: both footprints shrink further. Higher clouds are closer to the aircraft, so a fixed angular field covers less horizontal area.

For ASIM-like optics to include a compact patch at 5 km, the exit layer must be no higher than about **14.04 km toward a corner**, or **11.57 km along a side axis**, under this model.

How to handle the caption's 7.4 km distance

Only if 7.4 km is the *vertical distance to the same visible exit layer* may it be used as h. That hypothetical case gives FEGS offsets of 1.23–1.74 km and ASIM-like offsets of 4.39–6.21 km. A distance to the buried source, or a slant distance, is not automatically the required cloud clearance.

Why ordinary dimming is not the first limit here

As an illustration, take a small, horizontal, Lambertian exit patch and a horizontal receiver, without extra atmospheric loss. "Lambertian" means constant radiance with viewing direction. Simple geometry then gives

$$\frac{E(\rho)}{E(0)} = \left[1 + (\rho/h)^2\right]^{-2}. \tag{11}$$

At ASIM's diagonal edge this factor is about 0.344: a 50 mW/m^2 overhead reference would become about 17 mW/m^2. Under that particular model, the field stop excludes the patch before ordinary dimming makes it faint for an unattenuated ASIM-class sensor. An extended or anisotropic cloud halo need not obey this law.

8 What cloud scattering changes at an arbitrary offset

The relevant distinction is between the **position of the discharge** and the **positions of cloud regions illuminated by it**. A discharge outside the direct-view footprint may still light up cloud material inside it.

For a 15 km exit layer and a discharge 5 km ahead along a side axis, the halo must extend toward the aircraft by at least about **4.17 km for FEGS** or **2.03 km for ASIM-like optics** before it touches the respective footprint. Contact alone is not enough: a finite, sufficiently bright part must enter the accepted directions.

A millisecond arrival tail is not a map of this halo. Cloud-scattering simulations can predict spatial and temporal optical signatures,[6] but the supplied time curve alone does not determine that spatial distribution.

The actual calculation to perform

Let ρ be horizontal offset and ϕ its direction relative to a square footprint's axes. For a compact exit patch, its ideal geometric boundary is

$$\rho_{\mathrm{FOV}}(\phi) = \frac{h \tan \alpha}{\max(|\cos \phi|, |\sin \phi|)}. \tag{12}$$

This handles every azimuth, instead of using a single "maximum radius."

For a halo, use the arriving, band-integrated radiance $L_{337}(\Omega, t; \rho, \phi)$. Radiance describes how much light arrives *from each direction*; Ω labels direction on the sky. The accepted irradiance is

$$E_{\mathrm{acc}}(t; \rho, \phi) = \int_{\mathrm{FOV}} L_{337}(\Omega, t; \rho, \phi)\, a(\Omega) \cos\theta \, d\Omega. \tag{13}$$

Here $a(\Omega)$ is relative angular acceptance, and θ is the arrival angle from the aperture normal. Spectral throughput and window transmission must be included consistently, without duplicating factors already in A_{eff} or L_{337}.

The unsaturated ASIM-like signal in a chosen integration window T is

$$N_s = \frac{A_{\mathrm{eff}}}{\epsilon_\gamma} \int_T E_{\mathrm{acc}}(t)\, dt. \tag{14}$$

A basic illustration of the signal-to-noise ratio (SNR) is

$$\mathrm{SNR} \simeq \frac{N_s}{\sqrt{N_s + N_b + N_d + \sigma^2_{\mathrm{other}}}}, \tag{15}$$

where N_b and N_d are background and dark counts in the same window. The last term covers additional variance, including background estimation when applicable. The formula is not valid for severely saturated counting hardware.

What a genuine maximum-offset result looks like

Move the virtual detector through (ρ, ϕ), apply the instrument response and background/trigger model, and calculate detection probability or SNR. Only then can a chosen detection criterion define a maximum source offset. The threshold and azimuth must be stated; there is no universal range in kilometres supplied by the time trace alone.

9 Practical interpretation and the remaining checks

A faint halo may matter, but attenuation changes the trade-off

Relative to the plotted reference fluence, the unattenuated ASIM event benchmark corresponds to

$$f_{\text{bench}} = \frac{H_{\text{bench}}}{H_{\text{ref}}} \simeq 3.8 \times 10^{-5} \simeq 0.004\%.$$

This is a comparison of *accepted event energy*, not proof that a halo at any particular offset supplies it. The accepted fraction can vary with time; its fluence fraction need not equal its peak fraction.

If the required detected-count benchmark were held fixed, attenuation by 1000 would raise the corresponding reference-fluence fraction to about 3.8‰, and attenuation by 10,000 would raise it to about 38‰. Actual thresholds must be recalculated because attenuation also changes optical background. A bright-pulse operating range and maximum faint-halo sensitivity are therefore different design goals.

The minimum information needed to finish the offset assessment

Missing input	Why it matters
Original time arrays and normalization code	Resolve the apparent fluence difference and recover the unshown tail.
Source position, cloud-exit height and cloud geometry	Separate source distance from the height controlling the footprint.
Absolute radiance or photon arrivals versus position, direction and time	Determine how much of the halo actually enters each instrument at each offset.
Band response, angular response and aircraft/window geometry	Convert model output into accepted signal without double-counting loss factors.
Blue-channel calibration, background, operating range and detection rule	Determine a real detection probability, not just an ideal photon input.

Final assessment

ASIM-like at 20 km: a properly configured sensor should detect the plotted pulse easily when enough of the light is accepted and background is controlled. The original photon-counting operating range would be exceeded under full acceptance.
FEGS 337 nm: likely on brightness grounds when the illuminated region is in its central footprint, but a verified blue-channel calibration is needed for a quantitative detection claim.
Offsets: for an exit layer at 15 km, the compact-patch limits remain 0.83–1.18 km for FEGS and 2.97–4.20 km for ASIM-like optics. A 5 km compact patch is outside both. A scattered halo may permit detection of a more distant discharge; its maximum offset is not determined by the figure.

Changing pointing would change the geometric limits. It would not, by itself, establish the available irradiance at the newly viewed location. Likewise, these optical calculations neither predict an EFCM/LIP electric-field detection nor validate the source's electron or gamma-ray production.

A Check of the earlier FEGS red-channel comparison

This appendix preserves the useful numerical scale from the earlier answer while making its limitation explicit. **It is not the FEGS 337 nm sensitivity.**

Quick et al. report, for the 2017 *777 nm* data, a background RMS spectral radiance of 3.7×10^{-4} W m^{-2} sr^{-1} nm^{-1} and a six-RMS pulse criterion. Their radiance convention assumes a 1 nm effective lightning-emission bandwidth.[2]

For an ideal uniform square angular acceptance with side half-angle $\alpha = 9.46°$, the solid angle is

$$\Omega_{\mathrm{FOV}} = 4 \arctan\left(\frac{\tan^2 \alpha}{\sqrt{1 + 2\tan^2 \alpha}}\right) \simeq 0.108\,\mathrm{sr}. \tag{16}$$

A steradian (sr) is a unit of angular area. Approximating the projection factor by one over this small field gives the comparison

$$E_{\mathrm{bench},777} \simeq 6(3.7 \times 10^{-4})(1)(0.108) \tag{17}$$

$$\simeq 2.4 \times 10^{-4}\,\mathrm{W\,m^{-2}} = 0.24\,\mathrm{mW\,m^{-2}}. \tag{18}$$

The representative 50 mW/m^2 peak is about 210 times this idealized red-channel comparison. This calculation assumes the stated radiance convention and a simple uniform angular response, rather than a newly established blue-channel response.

Why it cannot be transferred directly to blue. The detector responsivity, optical/window transmission, calibration, gain, noise and saturation may differ. The 1 nm factor above is a red-emission convention, not the 10 nm physical filter width and not an appropriate universal width for 337 nm emission. The 2020 paper's numerical noise result does not independently calibrate the later blue configuration.

B Arithmetic, units and image-reading uncertainty

The constants used are $h_{\mathrm{P}} = 6.62607015 \times 10^{-34}$ J s, $c = 299\,792\,458$ m/s and $\lambda = 337.1 \times 10^{-9}$ m. Useful conversions are

$$1\,\mathrm{mW} = 10^{-3}\,\mathrm{W}, \quad 1\,\mu\mathrm{s} = 10^{-6}\,\mathrm{s}, \quad 1\,\mathrm{m^2} = 10^4\,\mathrm{cm^2}.$$

The reference photon fluence is

$$\frac{H_{\mathrm{ref}}}{\epsilon_\gamma} = 3.4 \times 10^{13}\,\mathrm{photons\,m^{-2}} = 3.4 \times 10^{9}\,\mathrm{photons\,cm^{-2}}.$$

For the image-area check, trace colours were identified away from the legend, the labelled axes supplied the scale, and the extracted curve was integrated with the trapezoidal rule. Pixel thickness, small hidden tails and missing samples limit the accuracy. The values were therefore rounded to about 2×10^{-5} and 6×10^{-6} J/m^2, not presented as precise measurements. The original simulation arrays remain the appropriate inputs for a final result.

The earlier Quick paper citation year should be **2020**, not 2021. The Fuglestad paper is **2026**, although its DOI contains the string "2025."

C Source figure and references

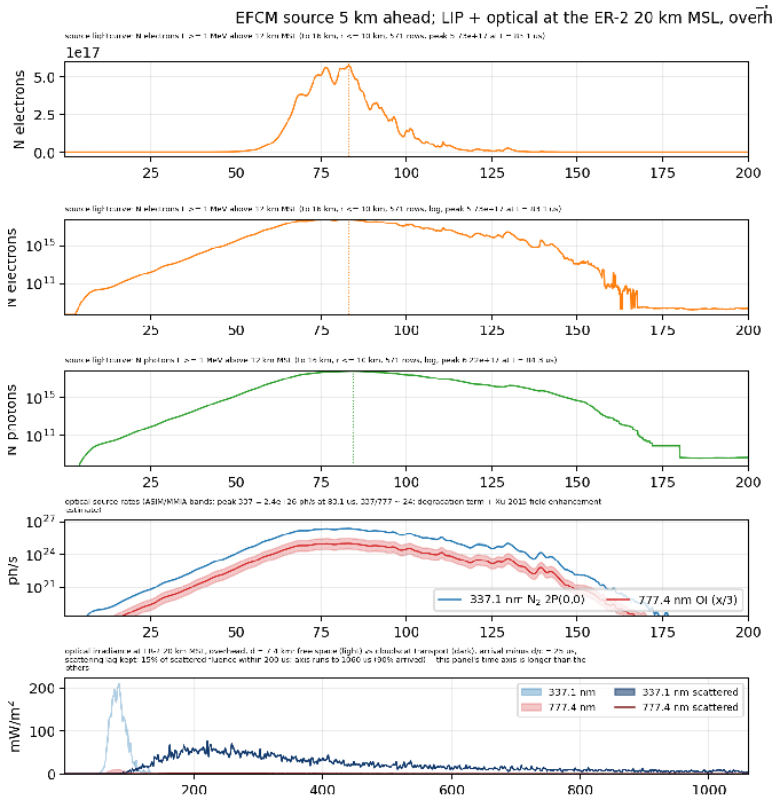


Figure 3: Full figure supplied by the user. This review assesses the bottom optical panel; it does not independently validate the physical model producing the upper panels.

References

[1] O. Chanrion et al. (2019). *The Modular Multispectral Imaging Array (MMIA) of the ASIM Payload on the International Space Station*. Space Science Reviews 215, 28. Tables 2, 5 and 6; Sections 3.3, 4.2 and 5.1. doi:10.1007/s11214-019-0593-y.

[2] M. G. Quick, H. J. Christian, K. S. Virts and R. J. Blakeslee (2020). *Airborne radiometric validation of the geostationary lightning mapper using the Fly's Eye GLM Simulator*. Journal of Applied Remote Sensing 14(4), 044518. Sections 2 and 4.3. doi:10.1117/1.JRS.14.044518.

[3] A. N. Fuglestad et al. (2026). *The Source Brightness Distribution of Terrestrial Gamma-Ray Flashes From the ALOFT Flight Campaign*. Journal of Geophysical Research: Atmospheres. Section 2, instrument description; published 11 March 2026. doi:10.1029/2025JD045235.

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[5] T. J. Lang et al. (2025). *Hunting for Gamma Rays above Thunderstorms: The ALOFT Campaign*. Bulletin of the American Meteorological Society 106(8), E1649–E1669. Sections 3b–3c, daytime blue observations and calibration context. doi:10.1175/BAMS-D-24-0060.1.

[6] A. Luque et al. (2020). *Modeling lightning observations from space-based platforms (CloudScat.jl 1.0)*. Geoscientific Model Development 13, 5549–5566. doi:10.5194/gmd-13-5549-2020.